Solutions to Exam (Tenta HT 23)
SF2940 Probability Theory

Problem 1

(a)

Let $A, B \in \mathcal{A}$[cite: 23]. Then

$$A \cup B = (A \cap B^c) \cup (A^c \cap B) \cup (A \cap B).$$

We observe that the events $(A \cap B^c)$, $(A^c \cap B)$, and $(A \cap B)$ are pairwise disjoint[cite: 23]. By the additivity of probability measures[cite: 23],

$$P(A \cup B) = P(A \cap B^c) + P(A^c \cap B) + P(A \cap B)[\text{cite: 23}].$$

Now we note that $(A \cap B^c) \cup (A \cap B) = A$. Since these events are disjoint, we have by additivity:

$$P(A \cap B^c) + P(A \cap B) = P(A) \implies P(A \cap B^c) = P(A) - P(A \cap B).$$

Similarly, $(A^c \cap B) \cup (A \cap B) = B$[cite: 23]. Since these events are disjoint, we have[cite: 23]:

$$P(A^c \cap B) + P(A \cap B) = P(B) \implies P(A^c \cap B) = P(B) - P(A \cap B)[\text{cite: 23}].$$

Substituting these into the identity above, it follows that[cite: 23]:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)[\text{cite: 23}].$$

(b)

Let Y be another random variable that has the same distribution as X, but is independent from X[cite: 23]. Then

$$\varphi_{X+Y}(t) = \varphi_X(t)\varphi_Y(t) = (\varphi_X(t))^2[\text{cite: 23}],$$

since X and Y have the same characteristic function[cite: 23].

(c)

For ease of notation, let $Y_i = \mathbf{1}_{\{X_i \leq x\}}$[cite: 23]. Since X_i are independent, so are Y_i[cite: 23]. The expected value of each Y_i is[cite: 23]

$$\mathbb{E}[Y_i] = \mathbb{E}[\mathbf{1}_{\{X_i \leq x\}}] = P(X_i \leq x) = F_{X_i}(x) =: F(x)[\text{cite: 23}].$$

Since X_i are identically distributed, we have[cite: 23]:

$$\mathbb{E}[\hat{F}_n(x)] = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[Y_i] = F_{X_1}(x) = F(x)[\text{cite: 23}].$$

Consider next[cite: 23]:

$$\mathbb{E}\left[(\hat{F}_n(x) - F(x))^2\right] = \mathbb{E}\left[\left(\frac{1}{n}\sum_{i=1}^{n}(Y_i - \mathbb{E}[Y_i])\right)^2\right] \text{[cite: 23]}$$
$$= \mathbb{E}\left[\frac{1}{n^2}\sum_{i=1}^{n}\sum_{k=1}^{n}(Y_i - \mathbb{E}[Y_i])(Y_k - \mathbb{E}[Y_k])\right] \text{[cite: 23]}$$
$$= \frac{1}{n^2}\sum_{i=1}^{n}\mathbb{E}\left[(Y_i - \mathbb{E}[Y_i])^2\right] + \frac{1}{n^2}\sum_{i\neq k}\mathbb{E}\left[(Y_i - \mathbb{E}[Y_i])(Y_k - \mathbb{E}[Y_k])\right] \text{[cite: 24].}$$

By independence, the covariance terms vanish[cite: 24]. Since the variables are identically distributed[cite: 24]:

$$\mathbb{E}\left[(\hat{F}_n(x) - F(x))^2\right] = \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(Y_i) = \frac{1}{n}\mathrm{Var}(Y_1) \text{[cite: 24].}$$

Yet the variance of Y_1 is bounded by one, as Y_1 takes values zero and one[cite: 24]:

$$\mathrm{Var}(Y_1) \leq \mathbb{E}[Y_1^2] = 0^2 \cdot P(Y_1 = 0) + 1^2 \cdot P(Y_1 = 1) \leq 1 \text{[cite: 24].}$$

Thus,

$$\mathbb{E}\left[(\hat{F}_n(x) - F(x))^2\right] \leq \frac{1}{n} \text{[cite: 24],}$$

and we have

$$\hat{F}_n(x) \xrightarrow[n\to\infty]{L^2} F(x) \text{[cite: 24],}$$

where $F(x)$ is the common cumulative distribution function of the $(X_i)_{i=1}^n$[cite: 24].

Alternative / Shortcut: (Y_i) are i.i.d. with Y_i taking values 0 and 1[cite: 24]. Hence, $\mathrm{Var}(Y_i) \leq 1$[cite: 24]. Indeed,

$$\mathrm{Var}(Y_i) = \mathbb{E}[Y_i^2] - (\mathbb{E}[Y_i])^2 = \mathbb{E}[Y_i] - (\mathbb{E}[Y_i])^2 = F(x) - F(x)^2 = F(x)(1 - F(x)) \leq \frac{1}{4} \leq 1 \text{[cite: 24],}$$

since $0 \leq F(x) \leq 1$[cite: 24]. By the weak law of large numbers for (Y_i)[cite: 24]:

$$\frac{1}{n}\sum_{i=1}^{n} Y_i \xrightarrow[n\to\infty]{\mathbb{P}} \mathbb{E}[Y_1] = F_{X_1}(x) = F(x) \text{[cite: 24].}$$

Problem 2

(a)

Let

$$B = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \text{[cite: 25]}.$$

Then

$$\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} = \begin{pmatrix} X_1 + X_3 \\ X_2 \end{pmatrix} = B \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix} \text{[cite: 25]}.$$

Since all X_i have zero expectation, the Y_i also have zero expectation[cite: 25]. The covariance matrix $\Lambda_{\mathbf{Y}}$ is, according to the linear transformation theorem, given by $\Lambda_{\mathbf{Y}} = B\Lambda_{\mathbf{X}}B^T$[cite: 25]. Thus,

$$\Lambda_{\mathbf{Y}} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 3 & 2 \\ 4 & 2 \end{pmatrix} = \begin{pmatrix} 6 & 3 \\ 3 & 2 \end{pmatrix} \text{[cite: 25]}.$$

Hence $\mathbf{Y} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 6 & 3 \\ 3 & 2 \end{pmatrix}\right)$[cite: 25].

Our goal is to compute[cite: 25]:

$$f_{Y_2|Y_1=y_1}(y_2) = \frac{f_{(Y_1,Y_2)}(y_1, y_2)}{f_{Y_1}(y_1)} \text{[cite: 25]}.$$

Here, we invert $\Lambda_{\mathbf{Y}}$ (via Cramer's rule)[cite: 25]:

$$\det(\Lambda_{\mathbf{Y}}) = 6 \cdot 2 - 3^2 = 3, \quad \Lambda_{\mathbf{Y}}^{-1} = \frac{1}{3}\begin{pmatrix} 2 & -3 \\ -3 & 6 \end{pmatrix} = \begin{pmatrix} \frac{2}{3} & -1 \\ -1 & 2 \end{pmatrix} \text{[cite: 25]}.$$

Thus, the joint density is[cite: 26]:

$$\begin{aligned} f_{(Y_1,Y_2)}(y_1, y_2) &= \frac{1}{2\pi\sqrt{3}} \exp\left(-\frac{1}{2}(y_1, y_2)\begin{pmatrix} \frac{2}{3} & -1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}\right) \text{[cite: 26]} \\ &= \frac{1}{2\pi\sqrt{3}} \exp\left(-\frac{1}{2}\left(\frac{2}{3}y_1^2 - 2y_1y_2 + 2y_2^2\right)\right) \text{[cite: 26]}. \end{aligned}$$

From above, we know that $Y_1 \sim \mathcal{N}(0, 6)$, hence[cite: 26]

$$f_{Y_1}(y_1) = \frac{1}{\sqrt{2\pi}\sqrt{6}} \exp\left(-\frac{1}{12}y_1^2\right) \text{[cite: 26]}.$$

Therefore,

$$\begin{aligned} f_{Y_2|Y_1=y_1}(y_2) &= \frac{\frac{1}{2\pi\sqrt{3}} \exp\left(-\frac{1}{3}y_1^2 + y_1y_2 - y_2^2\right)}{\frac{1}{\sqrt{2\pi}\sqrt{6}} \exp\left(-\frac{1}{12}y_1^2\right)} \text{[cite: 26]} \\ &= \frac{\sqrt{2}}{\sqrt{2\pi}} \exp\left(-\frac{1}{4}y_1^2 + y_1y_2 - y_2^2\right) \text{[cite: 26]} \\ &= \frac{1}{\sqrt{\pi}} \exp\left(-\left(y_2 - \frac{1}{2}y_1\right)^2\right) \text{[cite: 26]}. \end{aligned}$$

(c)

We introduce the variables[cite: 27]:

$$U = \frac{X_1}{X_2}, \quad V = X_2[\text{cite: 27}].$$

Then $X_1 = UV$ and $X_2 = V$[cite: 27]. The Jacobian of the transformation $(u, v) \mapsto (uv, v)$ is[cite: 27]

$$J = \det \begin{pmatrix} v & u \\ 0 & 1 \end{pmatrix} = v[\text{cite: 27}].$$

Thus, the joint density of (U, V) is[cite: 27]

$$f_{U,V}(u, v) = f_{X_1,X_2}(uv, v)|v|[\text{cite: 27}].$$

The covariance matrix of $(X_1, X_2)^T$ is $\Lambda = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$[cite: 27], with $\det(\Lambda) = 1$ and inverse[cite: 27]

$$\Lambda^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} [\text{cite: 27}].$$

Thus,

$$\begin{aligned} f_{X_1,X_2}(x_1, x_2) &= \frac{1}{2\pi} \exp\left(-\frac{1}{2}(x_1, x_2) \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) [\text{cite: 27}] \\ &= \frac{1}{2\pi} \exp\left(-\frac{1}{2}(2x_1^2 - 2x_1x_2 + x_2^2)\right) [\text{cite: 27}]. \end{aligned}$$

Setting $x_1 = uv$ and $x_2 = v$, we obtain[cite: 27]:

$$\begin{aligned} f_{X_1,X_2}(uv, v) &= \frac{1}{2\pi} \exp\left(-\frac{1}{2}(2u^2v^2 - 2uv^2 + v^2)\right) [\text{cite: 27}] \\ &= \frac{1}{2\pi} \exp\left(-\frac{v^2}{2}(2u^2 - 2u + 1)\right) [\text{cite: 27}]. \end{aligned}$$

Thus,

$$f_{U,V}(u, v) = \frac{1}{2\pi} \exp\left(-\frac{v^2}{2}(2u^2 - 2u + 1)\right) |v|[\text{cite: 28}].$$

Since $U = \frac{X_1}{X_2}$, we find the marginal distribution of U by integrating out v[cite: 28]:

$$\begin{aligned} f_U(u) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} |v| \exp\left(-\frac{v^2}{2}(2u^2 - 2u + 1)\right) dv[\text{cite: 28}] \\ &= \frac{1}{\pi} \int_0^{\infty} v \exp\left(-\frac{v^2}{2}(2u^2 - 2u + 1)\right) dv[\text{cite: 28}] \\ &= \frac{1}{\pi} \int_0^{\infty} v e^{-\frac{v^2}{2}a}\, dv[\text{cite: 28}], \end{aligned}$$

with $a := 2u^2 - 2u + 1$[cite: 28]. Note that

$$v e^{-\frac{v^2}{2}a} = -\frac{1}{a}\frac{\partial}{\partial v}\left(e^{-\frac{v^2}{2}a}\right) [\text{cite: 28}],$$

hence

$$\frac{1}{\pi} \int_0^{\infty} v e^{-\frac{v^2}{2}a}\, dv = -\frac{1}{\pi a}\left[e^{-\frac{v^2}{2}a}\right]_0^{\infty} = -\frac{1}{\pi a}(0 - 1) = \frac{1}{\pi a}[\text{cite: 28}].$$

Clearly, $a = u^2 - 2u + 1 + u^2 = (u-1)^2 + u^2 > 0$ for all $u \in \mathbb{R}$[cite: 28]. It follows that[cite: 29]:

$$f_U(u) = \frac{1}{\pi} \frac{1}{(u-1)^2 + u^2} = \frac{1}{\pi} \frac{1}{2u^2 - 2u + 1} = \frac{1}{\pi} \frac{\frac{1}{2}}{\left(u - \frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2}, \quad u \in \mathbb{R}[\text{cite: 29}],$$

which is the density of the Cauchy distribution $\mathcal{C}\left(\frac{1}{2}, \frac{1}{2}\right)$[cite: 29].

Problem 3

(a)

The probability generating function of X_1 (and of all other X_i) is[cite: 30]:

$$\begin{aligned} g_{X_1}(t) &= \sum_{k=1}^{\infty} t^k p_{X_1}(k) = \sum_{k=1}^{\infty} t^k \frac{m^k}{k!} \frac{1}{e^m - 1} \text{[cite: 30]} \\ &= \frac{1}{e^m - 1} \sum_{k=1}^{\infty} \frac{(tm)^k}{k!} = \frac{1}{e^m - 1} \left(\sum_{k=0}^{\infty} \frac{(tm)^k}{k!} - 1 \right) \text{[cite: 30]} \\ &= \frac{1}{e^m - 1}(e^{tm} - 1) = \frac{e^{-m}}{1 - e^{-m}}(e^{tm} - 1), \quad \text{for all } t \in \mathbb{R} \text{[cite: 30]}. \end{aligned}$$

(b)

Since N is independent of the (X_i)'s, we can use the composition formula for the probability generating function[cite: 30]:

$$g_{S_N}(t) = g_N(g_{X_1}(t)) \text{[cite: 30]}.$$

Since $p_N(k) = \binom{n}{k} p^k q^{n-k}$ for $k = 0, 1, \ldots, n$ with $p = 1 - e^{-m}$ and $q = e^{-m}$[cite: 30], we have:

$$g_N(t) = \sum_{k=0}^{n} \binom{n}{k} (tp)^k q^{n-k} = (q + tp)^n = (e^{-m} + t(1 - e^{-m}))^n \text{[cite: 30]}.$$

Thus,

$$\begin{aligned} g_{S_N}(t) &= \left(e^{-m} + g_{X_1}(t)(1 - e^{-m})\right)^n \text{[cite: 31]} \\ &= \left(e^{-m} + \frac{e^{-m}}{1 - e^{-m}}(e^{tm} - 1)(1 - e^{-m}) \right)^n \text{[cite: 31]} \\ &= (e^{-m} + e^{tm-m} - e^{-m})^n \text{[cite: 31]} \\ &= (e^{m(t-1)})^n \text{[cite: 31]} \\ &= e^{nm(t-1)}, \quad \text{for all } t \in \mathbb{R} \text{[cite: 31]}. \end{aligned}$$

This is the probability generating function of a Poisson random variable with parameter $n \cdot m$, i.e., $S_N \sim \text{Po}(n \cdot m)$ by the uniqueness theorem[cite: 31].

Problem 4

(a)

Since $Y \mid X = x \sim \text{Po}(x)$, its characteristic function is[cite: 32]:

$$\varphi_{Y|X=x}(t) = e^{x(e^{it}-1)}\text{[cite: 32]}.$$

By the law of total expectation, we have[cite: 32]:

$$\varphi_Y(t) = \mathbb{E}[e^{itY}] = \mathbb{E}[\mathbb{E}[e^{itY} \mid X]] = \mathbb{E}[e^{X(e^{it}-1)}]\text{[cite: 32]}.$$

Since $X \sim \Gamma(s, 1)$, using the moment generating function hint, we find[cite: 32]:

$$\varphi_Y(t) = \mathbb{E}[e^{(e^{it}-1)X}] = \frac{1}{(1-(e^{it}-1))^s} = \frac{1}{(2-e^{it})^s}\text{[cite: 32]}.$$

(b)

We first compute $\mathbb{E}[Y]$ and $\text{Var}(Y)$[cite: 32]. Note that

$$\frac{d}{dt}\varphi_Y(t) = \frac{-s}{(2-e^{it})^{s+1}} \cdot (-ie^{it}) = \frac{ise^{it}}{(2-e^{it})^{s+1}}\text{[cite: 32]}.$$

For the second derivative, we get[cite: 32]:

$$\frac{d^2}{dt^2}\varphi_Y(t) = \frac{-s(s+1)}{(2-e^{it})^{s+2}}(ie^{it})(ie^{it}) + \frac{is(ie^{it})}{(2-e^{it})^{s+1}} = \frac{s(s+1)e^{2it}}{(2-e^{it})^{s+2}} - \frac{se^{it}}{(2-e^{it})^{s+1}}\text{[cite: 32]}.$$

Evaluating at $t = 0$[cite: 32]:

$$\varphi_Y'(0) = is, \quad \varphi_Y''(0) = s(s+1) - s = -s(s+1) - s = -(s^2+2s)\text{[cite: 32]}.$$

Thus,

$$\mathbb{E}[Y] = s, \quad \mathbb{E}[Y^2] = s^2 + 2s, \quad \text{Var}(Y) = 2s\text{[cite: 33]}.$$

Set

$$Z := \frac{Y - \mathbb{E}[Y]}{\sqrt{\text{Var}(Y)}} = \frac{Y-s}{\sqrt{2s}}\text{[cite: 33]}.$$

Then its characteristic function is[cite: 33]:

$$\varphi_Z(t) = \mathbb{E}\left[\exp\left(i\frac{t}{\sqrt{2s}}(Y-s)\right)\right] = e^{-it\frac{s}{\sqrt{2s}}}\mathbb{E}\left[e^{i\frac{t}{\sqrt{2s}}Y}\right] = e^{-it\sqrt{\frac{s}{2}}}\varphi_Y\left(\frac{t}{\sqrt{2s}}\right)\text{[cite: 33]}.$$

By part (a), we have[cite: 33]:

$$\varphi_Z(t) = e^{-it\sqrt{\frac{s}{2}}}\frac{1}{\left(2-e^{i\frac{t}{\sqrt{2s}}}\right)^s} = e^{-it\sqrt{\frac{s}{2}}}\frac{1}{\left(1-\left(e^{i\frac{t}{\sqrt{2s}}}-1\right)\right)^s}\text{[cite: 33]}.$$

We analyze the asymptotic behavior as $s \to \infty$ by considering the logarithm[cite: 33]:

$$\log\left(\frac{1}{\left(1-\left(e^{i\frac{t}{\sqrt{2s}}}-1\right)\right)^s}\right) = -s\log\left(1-\left(e^{i\frac{t}{\sqrt{2s}}}-1\right)\right)\text{[cite: 33]}.$$

As $s \to \infty$, expanding the exponential function gives[cite: 33]:

$$e^{i\frac{t}{\sqrt{2s}}} - 1 = \frac{it}{\sqrt{2s}} - \frac{t^2}{4s} + o\left(\frac{1}{s}\right) \text{[cite: 33]}.$$

Using $\log(1-u) = -u - \frac{u^2}{2} + o(u^2)$ with $u = \frac{it}{\sqrt{2s}} - \frac{t^2}{4s} + o\left(\frac{1}{s}\right)$[cite: 33]:

$$\begin{aligned}
-s\log\left(1 - \left(e^{i\frac{t}{\sqrt{2s}}} - 1\right)\right) &= -s\left(-\left(\frac{it}{\sqrt{2s}} - \frac{t^2}{4s}\right) - \frac{1}{2}\left(\frac{it}{\sqrt{2s}}\right)^2 + o\left(\frac{1}{s}\right)\right) \text{[cite: 33]} \\
&= -s\left(-\frac{it}{\sqrt{2s}} + \frac{t^2}{4s} + \frac{t^2}{4s} + o\left(\frac{1}{s}\right)\right) \\
&= -s\left(-\frac{it}{\sqrt{2s}} + \frac{t^2}{2s} + o\left(\frac{1}{s}\right)\right) \\
&= it\sqrt{\frac{s}{2}} - \frac{t^2}{2} + o(1) \quad \text{as } s \to \infty.
\end{aligned}$$

Returning to the characteristic function, we obtain:

$$\varphi_Z(t) = e^{-it\sqrt{\frac{s}{2}}} \cdot \exp\left(it\sqrt{\frac{s}{2}} - \frac{t^2}{2} + o(1)\right) = e^{-\frac{t^2}{2} + o(1)}.$$

Since the exponential function is continuous,

$$\lim_{s\to\infty} \varphi_Z(t) = e^{-\frac{1}{2}t^2}.$$

By the continuity theorem (uniqueness theorem), we conclude that

$$\frac{Y - \mathbb{E}[Y]}{\sqrt{\mathrm{Var}(Y)}} \xrightarrow[s\to\infty]{d} \mathcal{N}(0,1).$$